

The background features a minimalist design with three orange circles of varying sizes on the left side. A thin orange curved line starts near the top circle and extends towards the center. Another thin orange curved line starts near the bottom circle and extends towards the center. On the right side, there is a large orange circle partially visible, and a thin orange curved line starts near it and extends towards the center. The text 'FACE DETECTION' is centered in a bold, dark blue, sans-serif font.

FACE DETECTION

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Smart Attendance System using Face Recognition in AI

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Abstract

This project aims to automate attendance using face detection technology.

It captures facial data in real-time via webcam, compares it with pre-stored data, and marks attendance automatically.

This system improves accuracy, reduces proxy attendance, and saves manual effort.

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Introduction

Manual attendance systems are time-consuming and prone to errors or manipulation.

The use of computer vision and AI enables contactless, quick, and reliable attendance marking.

This project uses Python, OpenCV, and face_recognition libraries for implementation.

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Existing System

Traditional methods:

- Manual attendance via register or fingerprint scanner.

Digital methods:

- Barcode or RFID-based attendance.

Limitations:

- Time-consuming.
- Possibility of proxy attendance.
- Lack of real-time verification.

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Disadvantages of Existing System

- **Easy to manipulate (proxy attendance)**
- **Time wastage during roll calls**
- **Physical contact in fingerprint-based systems (unhygienic)**
- **No automatic tracking or data logs**

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Proposed System

- **Uses real-time face recognition to detect and recognize individuals.**
- **Automatically updates attendance in a database.**
- **Ensures single-entry per day using timestamps.**
- **Provides logs for admin to verify who attended and when.**

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Software Requirements

Operating System

- Windows 10 or higher

Libraries and Frameworks

- OpenCV (Face Detection)
- Tkinter (GUI Design)
- NumPy (Image Processing)
- Pandas (CSV Handling)
- PIL (Image Rendering)
- smtplib, email.mime (Email Integration)
- playsound (Audio Feedback)
- glob and os (File Management)

Programming Language

- Python 3.8 or Higher

Additional Tools

- Haarcascade XML (Face Detection)
- Microsoft Outlook or Gmail (SMTP Mail Configuration)

IDLE / Editor

- Python's built-in editor

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Hardware Requirements

Processor:

- Intel Core i3 / Ryzen 3 or higher

Display:

- 1280x720 resolution or higher (Full HD preferred)

Storage:

- At least 500 MB of free space

Camera:

- HD Webcam (Built-in or External USB)

Microphone (Optional):

- For voice-based commands (future upgrade scope)

RAM:

- Minimum 4 GB (Recommended: 8 GB)

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Advantages

- **Contactless and hygienic**
- **Reduces human error and manipulation**
- **Time-efficient**
- **Accurate logging and reporting**
- **Integration with database and GUI possible**

Literature Review



1. M. Turk and A. Pentland, “Eigenfaces for recognition” – introduced face recognition using PCA.
2. Dlib & HOG algorithm – widely used for fast and accurate face location.
3. Recent advancements include deep learning methods (CNNs) for high accuracy.

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Conclusion

The face detection attendance system provides a smart, reliable, and efficient alternative to traditional methods.

By automating the process, it eliminates manual errors, improves security, and brings AI into daily use in educational or corporate environments

References

[1] <https://pypi.org/project/face-recognition/>

[2] <https://opencv.org/>

[3] M. Turk and A. Pentland (1991), “Face recognition using Eigenfaces”

[4] Dlib Library Documentation

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***Thanks for giving wonderful
opportunity to us***

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